Holonomic Gauge Theory and Singer's Path Spaces

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What is Gauge Theory?

. . . depends on who you ask! (kind of)

Gauge Theory in Physics

The study of symmetries or invariances of the dynamics of a given physical system (for example, physical symmetries of the electromagnetic force).

Gauge Theory in Mathematics

The study of connections and their curvature on principal fiber bundles.

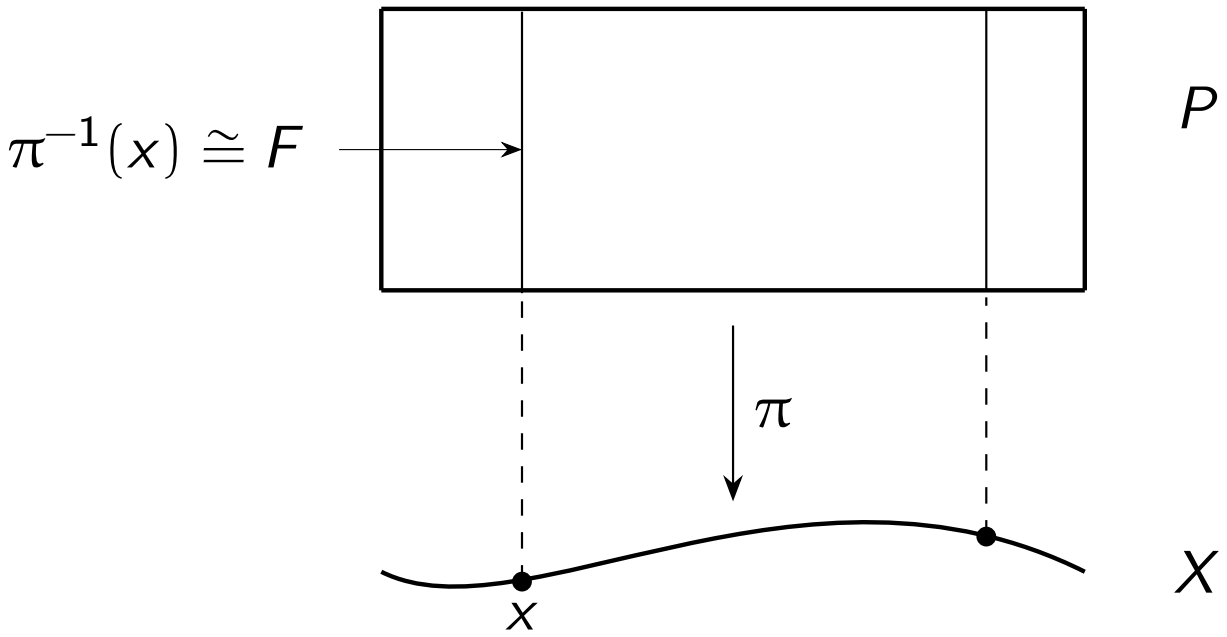
Simply Gauge Theory

Physicists and mathematicians were studying the *very same object*!

Fiber Bundles

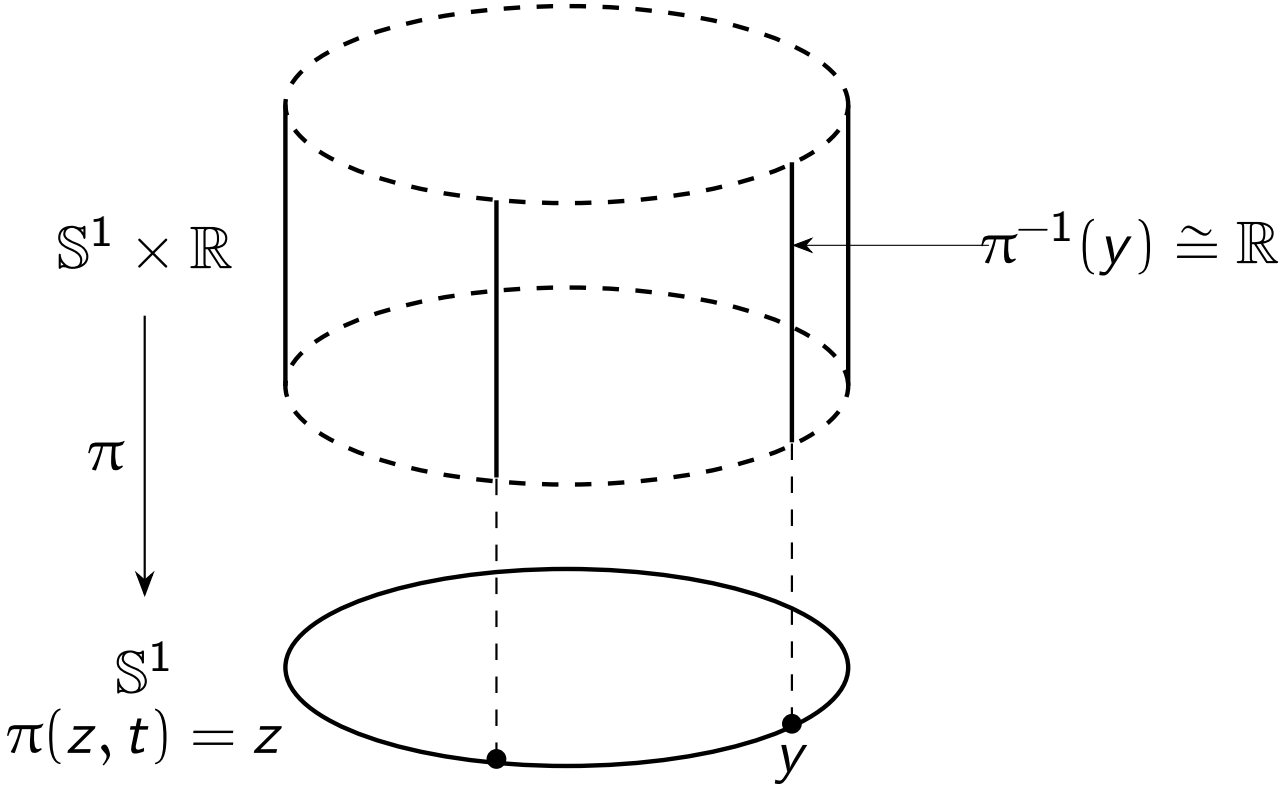
Definition: Fiber Bundles

A fiber bundle is a surjection $\pi : P \to X$ of spaces (say, manifolds) such that P locally looks like $X \times F$ for some space F. Each $\pi^{-1}(x) \subseteq P$ is called a fiber over x and is an isomorphic copy of F.



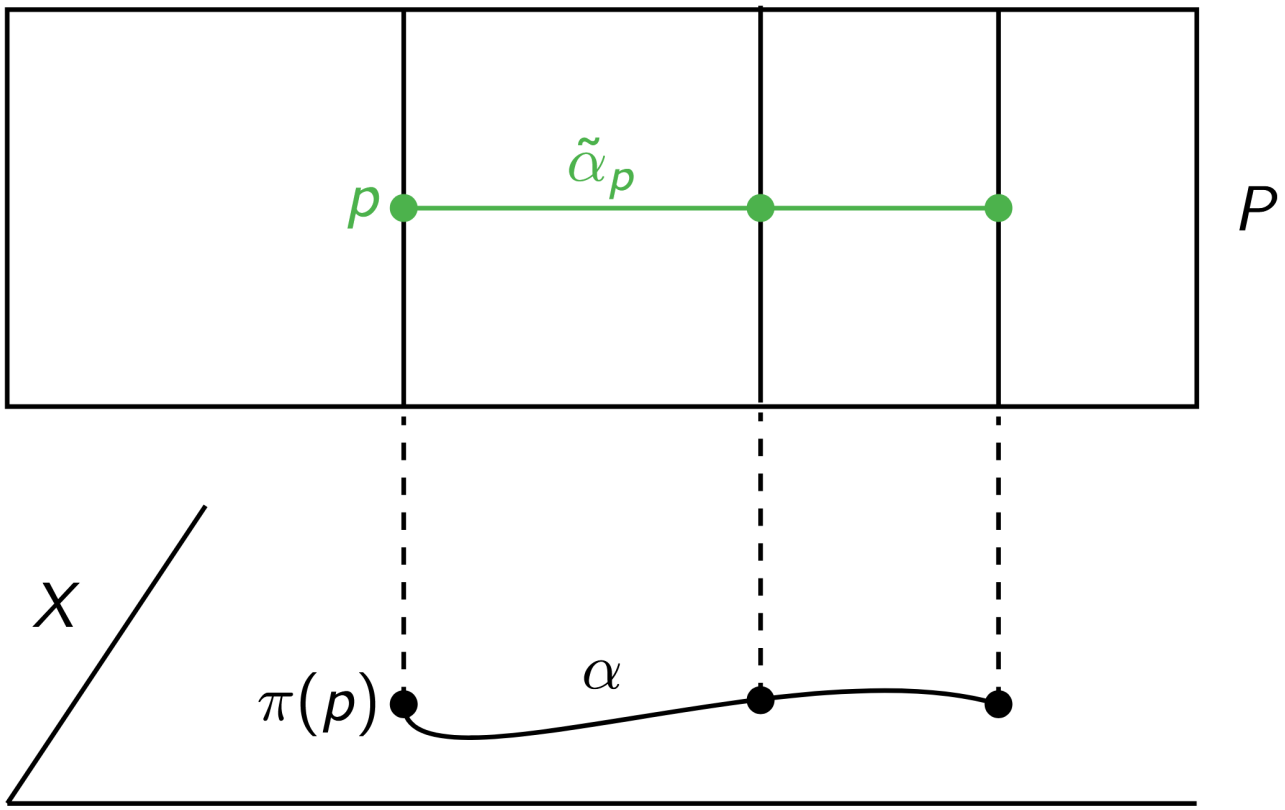
Principal Bundles

We will be interested in the case when $F = G$ is a group which acts on each fiber. Such a fiber bundle $\pi : P \to X$ is called a principal G-bundle.



Parallel Transport

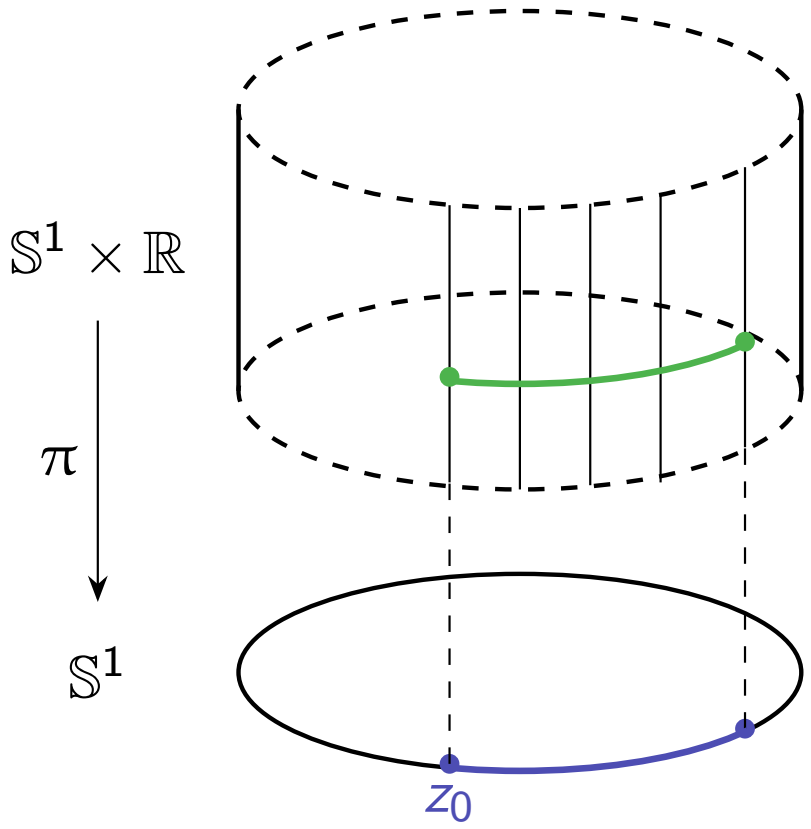
On every principal bundle there is a mechanism, called parallel transport or connection, to lift paths $\alpha : [0,1] \to X$ to paths $\tilde{\alpha}_p : [0,1] \to P$ starting at any $p \in \pi^{-1}(\alpha(0))$ on P "horizontally."



These horizontal lifts are unique (after choosing the starting point p).

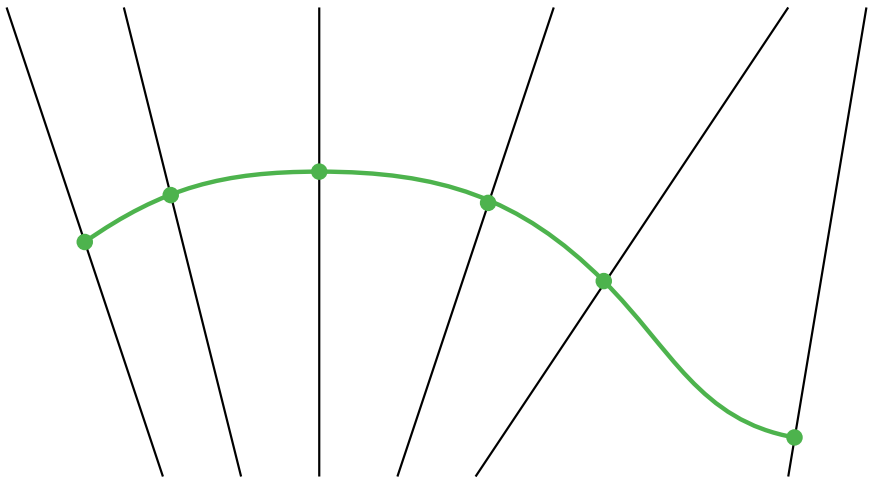
Example: Parallel Transport for the Cylinder

Given a path $\alpha : [0,1] \to \mathbb{S}^1$ starting at $z_0 \in \mathbb{S}^1$, the path $\tilde{\alpha}_{(z_0,x_0)} : [0,1] \to \mathbb{S}^1 \times \mathbb{R}$ given by $\tilde{\alpha}_{(z_0,x_0)}(t) = (\alpha(t), x_0)$ is a horizontal lift:



Parallel Transport in Nontrivial Bundle

Bundles are only required to be locally trivial—they might have curvature:



Forces as Curvature of a Connections

It turns out the fundamental forces of physics can be described as the curvature of a connection on principal G-bundles.

Example: Electromagnetism

Maxwell's four equations for electromagnetism arise from the curvature of a connection on a $U(1)$-bundle $\pi : P \to M$, where M is Minkowski space ($M = \mathbb{R}^4$ with the Lorentz metric).

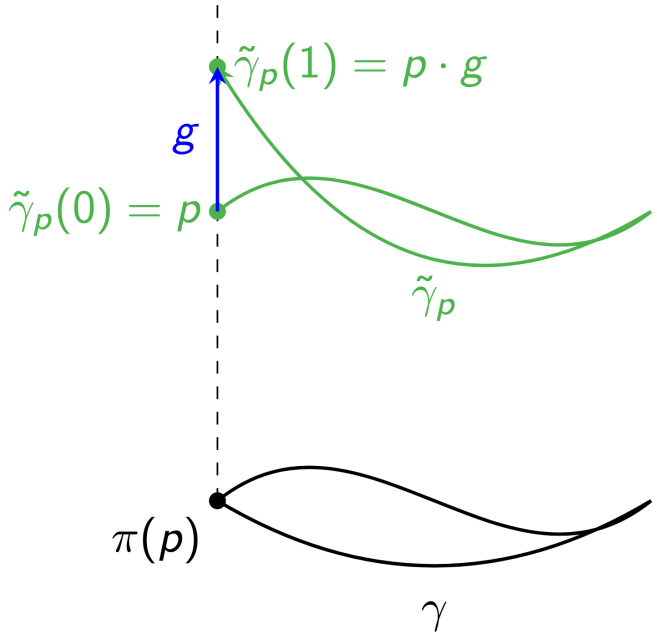
Holonomy and Curvature

Gauge theory studies connections and their curvature.

One approach is to look at the holonomy of a given connection, which can be called holonomic gauge theory.

Parallel Transport and Holonomy

Let $\gamma : [0,1] \to M$ be a loop. When we horizontally lift γ to $\tilde{\gamma}_p$, we cannot guarantee that $\tilde{\gamma}_p$ is a loop:



In general, $\tilde{\gamma}_p(1) = \tilde{\gamma}_p(0) \cdot g = p \cdot g$ for some $g \in G$.

Definition: Holonomy

The holonomy at $p \in P$ (of a connection on $\pi : P \to X$) is the subgroup $\mathrm{Hol}_p \leq G$ of $g \in G$ where there's a loop γ in X with $\tilde{\gamma}_p(1) = p \cdot g$.

Group Structures

Fix any point $\bullet \in M$ and denote by ΩM the group of loops based at $\bullet$.

Since $\mathrm{Hol}_p \leq G$ and ΩM are groups, we expect an algebraic relationship between them.

Definition:

The holonomy map (based at $p \in P$ for a given connection on $\pi : P \to M$) is the map $H : \Omega M \to G$ defined implicitly by the equation

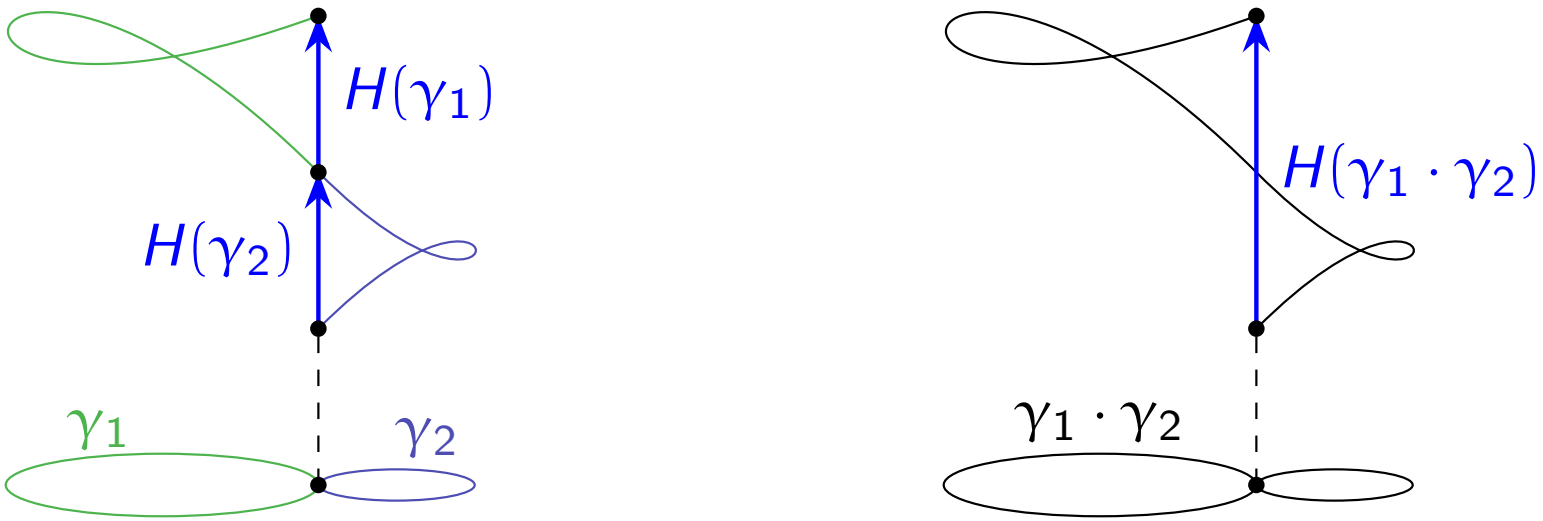
$$\tilde{\gamma}_p(1) = p \cdot H(\gamma), \quad \text{for all } \gamma \in \Omega M.$$

Holonomy Maps are Homomorphisms

Holonomy Maps are Homomorphisms

Given a connection on $\pi : P \to M$, the holonomy map $H : \Omega M \to G$ given by $\tilde{\gamma}_p(1) = p \cdot H(\gamma)$ for all $\gamma \in \Omega M$ is a group homomorphism. Also,

$$\mathrm{Hol}_p = H(\Omega M).$$



Homomorphisms are Holonomy Maps

It is maybe a bit surprising that the converse is true...

Theorem (Kobayashi 1954)

Fix a space M. For any Lie group G and homomorphism $H : \Omega M \to G$, there exists a principal G-bundle over M with a connection such that

$$\mathrm{Hol} = H(\Omega M).$$

We outline an argument due to I.M. Singer (Singer 1995).

Singer's Path Spaces

Fix a space M and any point $\bullet \in M$.

Space of Based Paths

Let $\mathcal{F}$ be the set of paths $\alpha : [0,1] \to M$ starting at $\bullet$, i.e. $\alpha(0) = \bullet$ and $\alpha(1) \in M$ can be arbitrary.

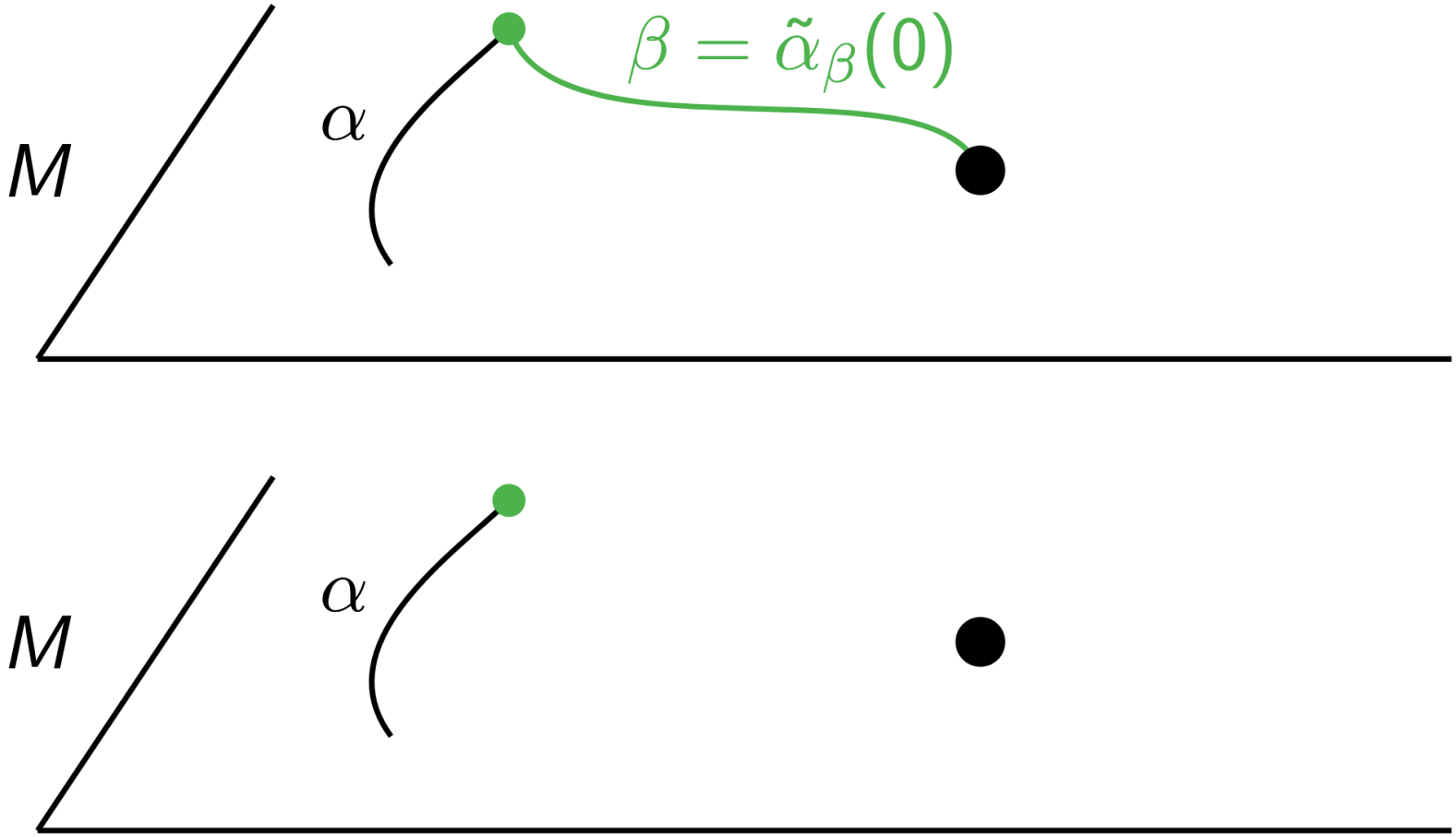
There is a natural "target map" $\tau : \mathcal{F} \to M$ given by

$$\tau(\alpha) = \alpha(1).$$

Note that $\tau : \mathcal{F} \to M$ looks like a principal ΩM-bundle.

The Universal Connection

We have a notion of parallel transport on $\tau : \mathcal{F} \to M$, which Singer calls the universal connection: for a path $\alpha : [0,1] \to M$ and an initial "point" $\beta \in \mathcal{F}$, we define $\tilde{\alpha}_\beta : [0,1] \to \mathcal{F}$ by...



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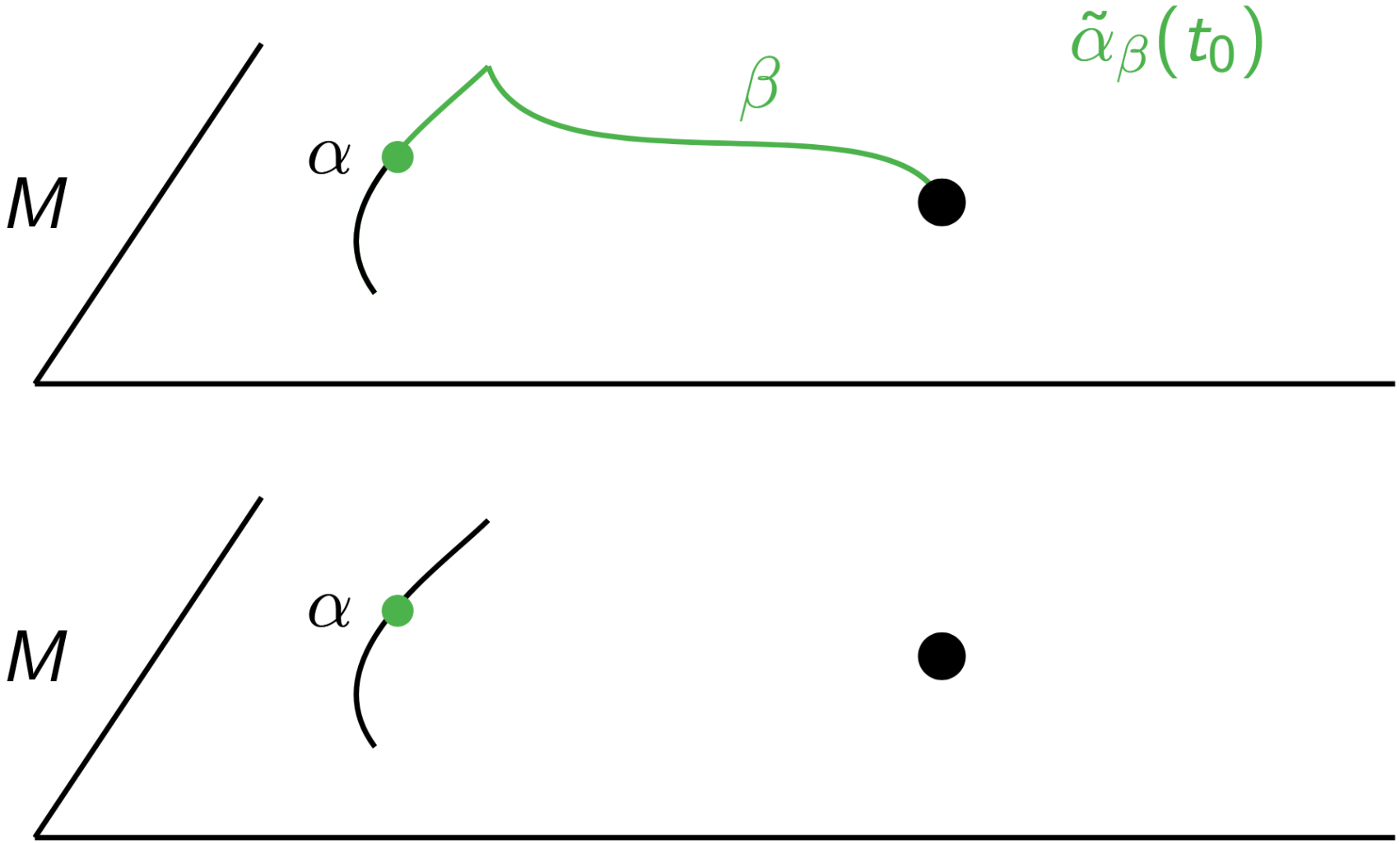

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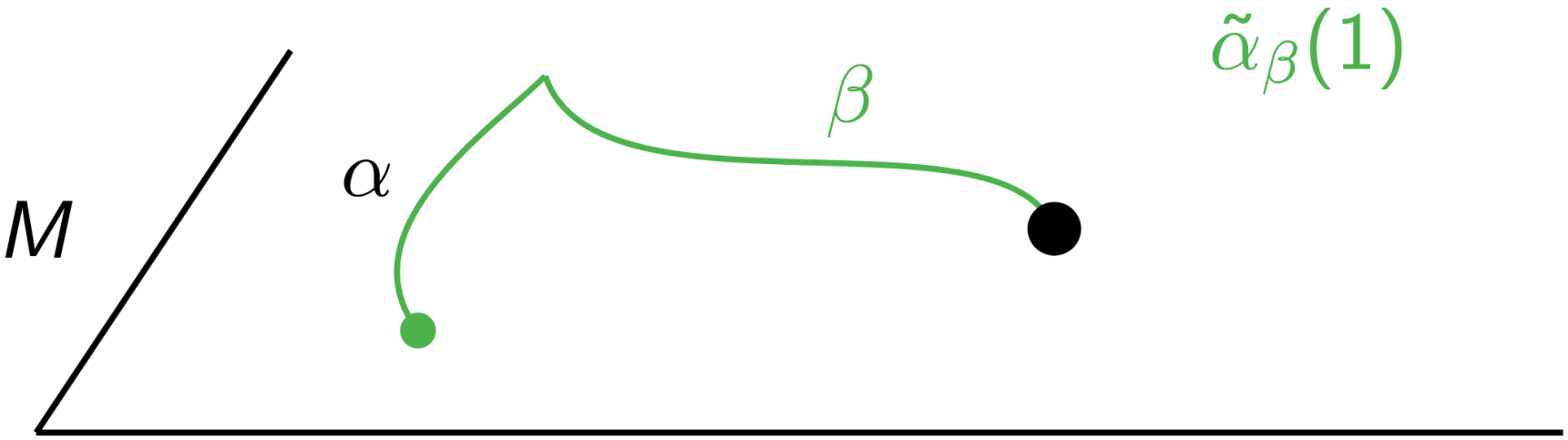

Induced Horizontal Lift on Associated Bundle

We have an ΩM-bundle, but there is a natural way to obtain a principal G-bundle over M by constructing an associated bundle:

Associated Bundle

The associated bundle $\tau_H : \mathcal{F} \times_H G \to M$ is a principal G-bundle. Elements of $\mathcal{F} \times_H G$ are equivalence classes of $\mathcal{F} \times G$,

$$[\alpha, g] = \{(\alpha \cdot \gamma, H(\gamma)^{-1} g) : \gamma \in \Omega M\},$$

and $\tau_H([\alpha, g]) = \tau(\alpha)$ with action $[\alpha, g_1]g_2 = [\alpha, g_1 g_2]$.

Induced Horizontal Lift

The horizontal lift of $\alpha : [0,1] \to M$ at $[\beta, g] \in \mathcal{F} \times_H G$ is the path

$$s \mapsto [\tilde{\alpha}_\beta(s), g]$$

Computing the Holonomy

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- Let $\gamma : [0,1] \to M$ be a loop at $\bullet$ in M.
- Horizontally lift, starting at $[\bullet, e] \in \mathcal{F} \times_H G$.
- The horizontal lift $s \mapsto [\tilde{\gamma}_\bullet(s), e]$ ends at $(s = 1)$:

$$[\tilde{\gamma}_\bullet(1), e] = [\gamma, e] = [\gamma \cdot \gamma^{-1}, H(\gamma)e] = [\bullet, e] \cdot H(\gamma).$$

Theorem (Kobayashi 1954)

For any Lie group G and homomorphism $H : \Omega M \to G$, there exists a principal G-bundle over M with a connection such that

$$\mathrm{Hol} = \{H(\gamma) : \gamma \in \Omega M\}.$$

References

- Kobayashi, Shôshichi (1954). "La connexion des variétés fibrées. II". In: *C. R. Acad. Sci. Paris* 238, pp. 443–444.
- Singer, Isadore Manuel (1995). "On the Master Field in Two Dimensions". In: *Functional Analysis on the Eve of the 21st Century*. Ed. by Simon Gindikin, James Lepowsky, and Robert Wilson. Birkhäuser, pp. 263–281.